Deepali Mondal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.16
GVN The Global School Bhopal	Class XII (CBSE)	2023	88.20%
GVN The Global School Bhopal	Class X (CBSE)	2021	95.60%

RESEARCH EXPERIENCE & PROJECTS

Zebrafish Neurobehavioral Toxicology - Glyphosate & Triclosan | Cell Signalling and Ion Channel Biology Lab, IITH

- Statistically analysed existing zebrafish behavioural data on glyphosate exposure to test a sex-specific response hypothesis; found no significant male-female difference
- Gained exposure to wet-lab techniques used in the study - dissection, brain lysate extraction, ELISA, PCR
- Currently contributing to ongoing work on the mechanistic basis of Roundup-induced anxiety/recovery, compared against triclosan exposure

Doxorubicin Sensitivity Prediction from Gene Expression (NCI-60)

- Built an end-to-end ML pipeline to predict cancer cell line sensitivity to doxorubicin from gene expression, designed around the NCI's public CellMiner database
- Implemented feature selection (top-variance genes), Random Forest regression, and 5-fold cross-validation; validated the full pipeline using synthetic data with an injected signal
- Designed the pipeline to recover known doxorubicin biology (e.g. efflux transporter ABCB1, drug target TOP2A) as a built-in sanity check for biological plausibility
- Built with Python, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn in Jupyter Notebook

Germline Variant-Based Cardiotoxicity Risk Prediction

- Curated a 14-variant pharmacogenomic panel from PharmGKB and GWAS literature spanning 9 genes across drug transport, metabolism, and oxidative stress pathways
- Trained a Random Forest classifier (ROC-AUC 0.66) on genotype-simulated cohort (N=500) with outcomes derived from published odds ratios
- SHAP analysis recovered RARG and UGT1A6 as top risk drivers, consistent with Nature Genetics GWAS findings

SKILLS

Lab Techniques: PCR, Agarose Gel Electrophoresis, SDS-PAGE, Western Blotting, ELISA, DNA/RNA Extraction & Quantification, Bacterial Transformation, Microbial/Animal/Plant Cell Culture

Bioinformatics Tools: Biopython, BLAST, MATLAB

Programming Languages: Python, R, C, C++

Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn

Tools & Platforms: Git, GitHub, Linux, Bash/Shell

Databases: SQL, NCBI Tools, UCSC Genome Browser

RELEVANT COURSES

Core Biology: Biochemistry, Microbiology, Molecular & Cellular Biology, Protein Structure, Function and Disease

Bioinformatics: Basic Bioinformatics, Next Generation Sequencing, Nucleic Acids and RNA Biomedicine, Molecular Biophysics and Macromolecular Structural Biology

Computational: Data Structures & Applications, R and Python for Biologists, Big Data Biology & Biological Databases, Biostatistics

POSITIONS OF RESPONSIBILITY

Head of Public Relations | International Relations Cell, IIT Hyderabad

- Lead the PR function of IR Cell, overseeing outreach/social media/branding for the cell's international engagement initiatives

- Coordinate with team members to manage public communication around exchange programs, scholarship databases, and international student events

Outbound and PR Coordinator | International Relations Cell, IIT Hyderabad

- Promote outbound opportunities -- study abroad programs, exchange semesters, and scholarship listings -- to the student community

- Manage PR/social media presence to publicise IR Cell initiatives and partner-university opportunities

Inbound Coordinator | International Relations Cell, IIT Hyderabad

- Coordinated hosting and onboarding of incoming international students/exchange scholars on campus

- Organised Buddy Program/orientation events to help international students integrate into campus life

Events Manager | E-Cell, IIT Hyderabad

- Managed planning and execution of E-Cell events such as E-Summit / Startup Fair/workshops, coordinating logistics, speakers, and sponsors

- Liaised with VCs/startups/sponsors to facilitate event partnerships

SCHOLASTIC ACHIEVEMENTS

Qualified for the **PRMO (**now **IOQM)** exam

EXTRACURRICULAR

National Service Scheme (NSS), Member, IIT Hyderabad

- Participated in blood donation camps, campus/community cleanliness drives, and tree plantation initiatives